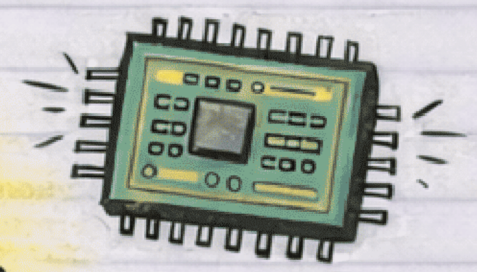
COA Module-4 Notes



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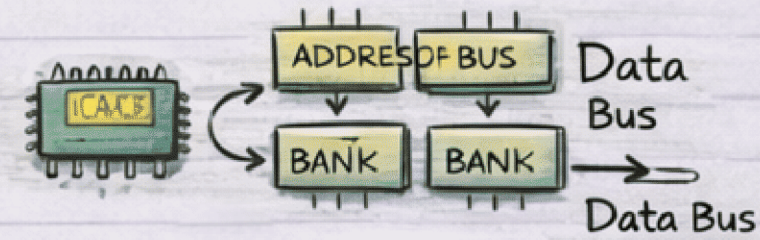
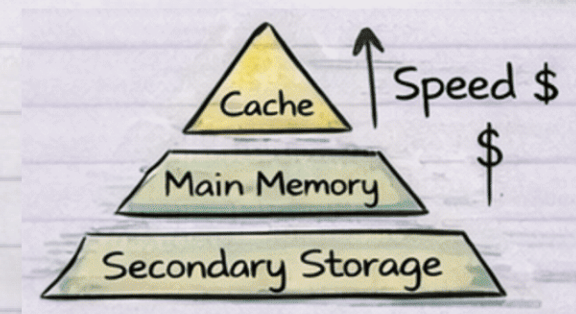
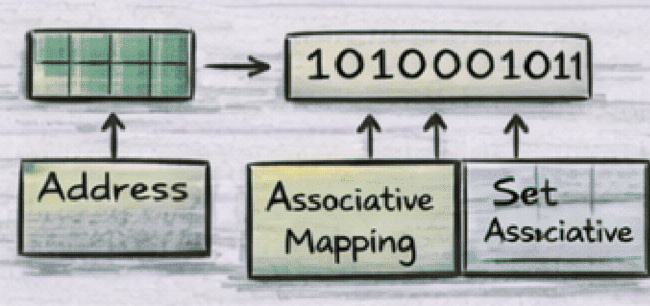
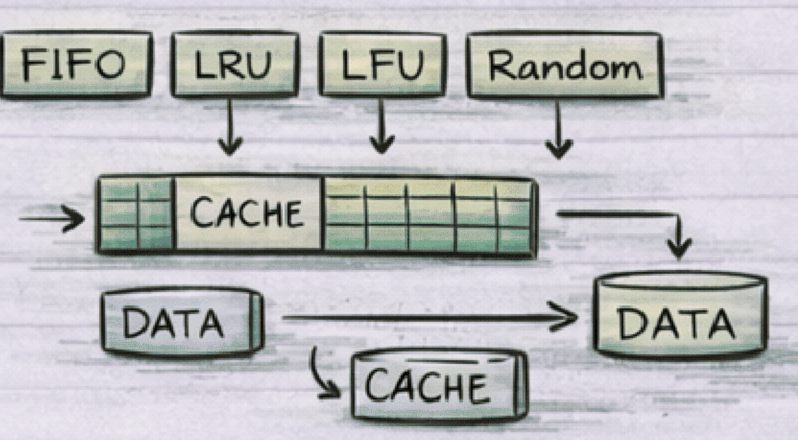
Contents Covered:

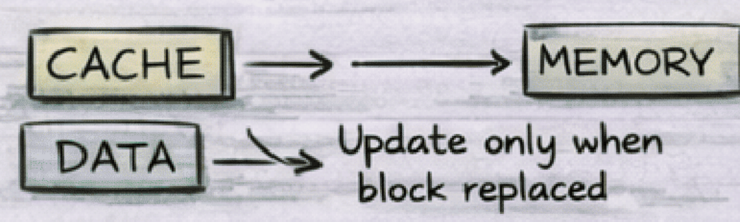
- Intro to Memory Interleaving.
- Hierarchical Memory Organization.
- Cache Memory.
- Mapping Functions
- Replacement Algorithms
- Write Policies
  - Write-through Method
  - Write-back Method



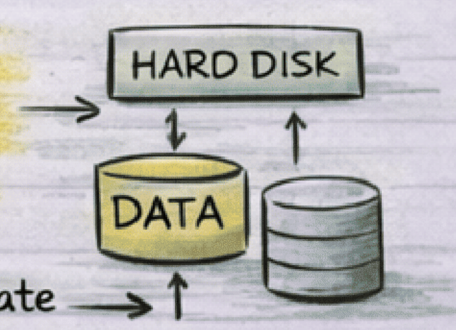








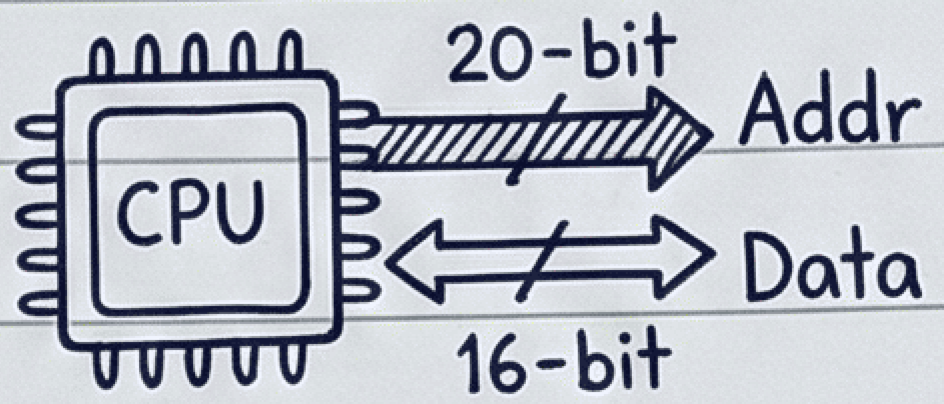
- Advantage of Write-through Cache Policy
  - Immediate Update



Module 4: Memory Organization - Memory Interleaving (8086)

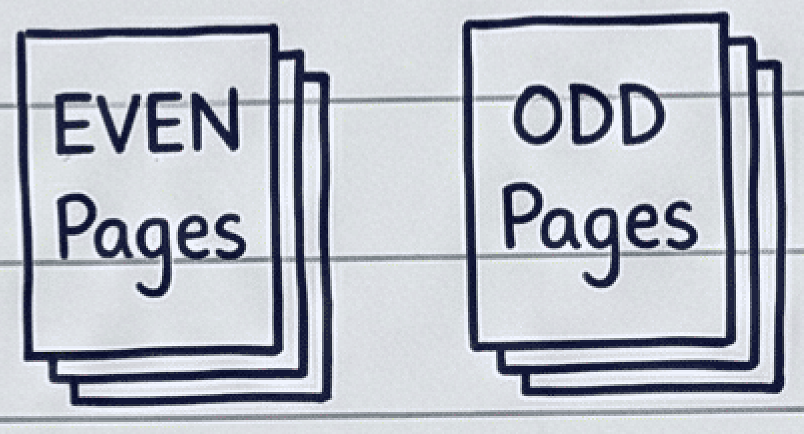
CONTEXT (The Problem):

- 8086 has a 20-bit address bus but only a 16-bit data bus.
- Goal: Speed up access!



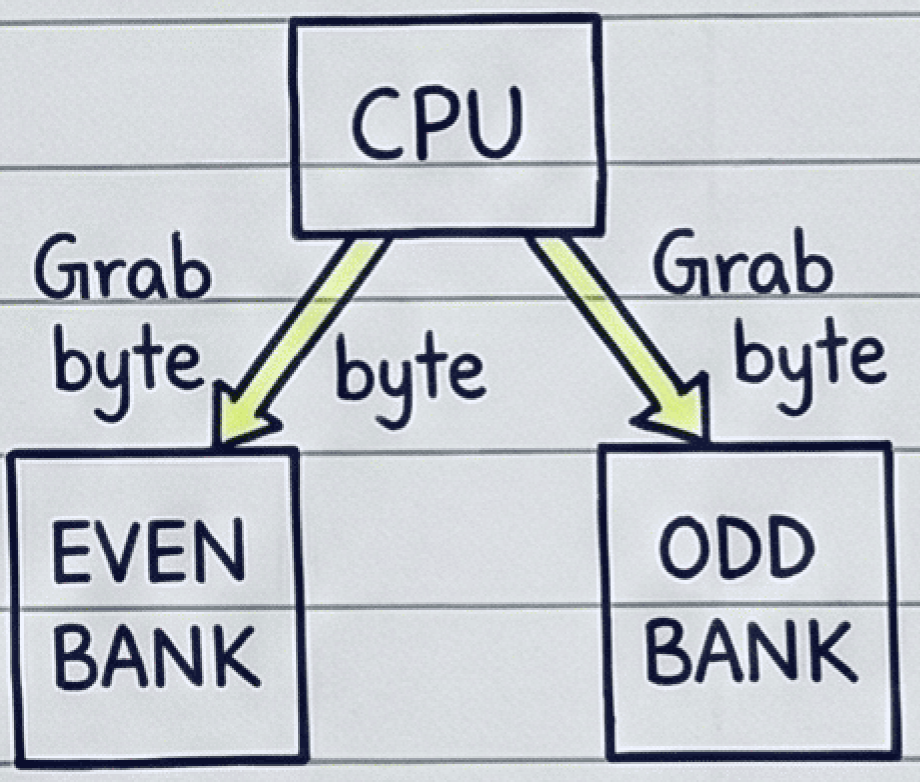
THE SOLUTION: Interleaving

- Divide memory into Two Banks: One for Even addresses, one for Odd addresses.



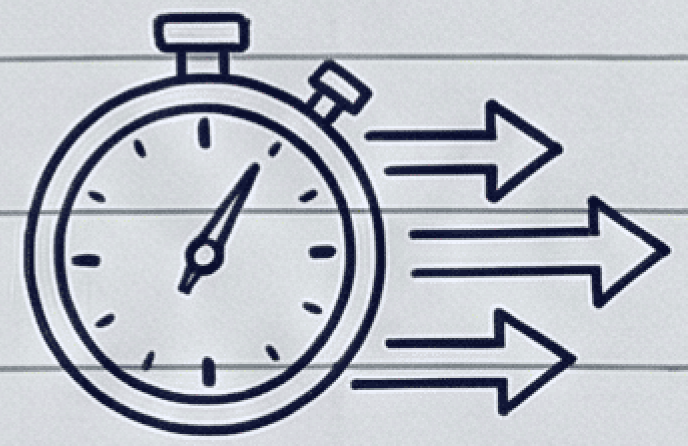
HOW IT WORKS:

- CPU can fetch a 16-bit word (two bytes, one even, one odd) in a Single cycle by accessing both banks Simultaneously.



BENEFITS:

- Concurrency: While one bank is busy, the other can start.
- Faster fetching than single-byte access.
- Increased Bandwidth (data transfer rate).
- Great for Pipelined processors.



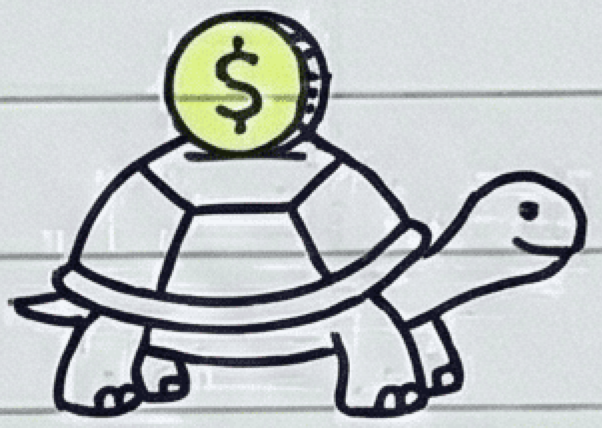
Concept of Hierarchical Memory Organization

THE NEED (The Problem):

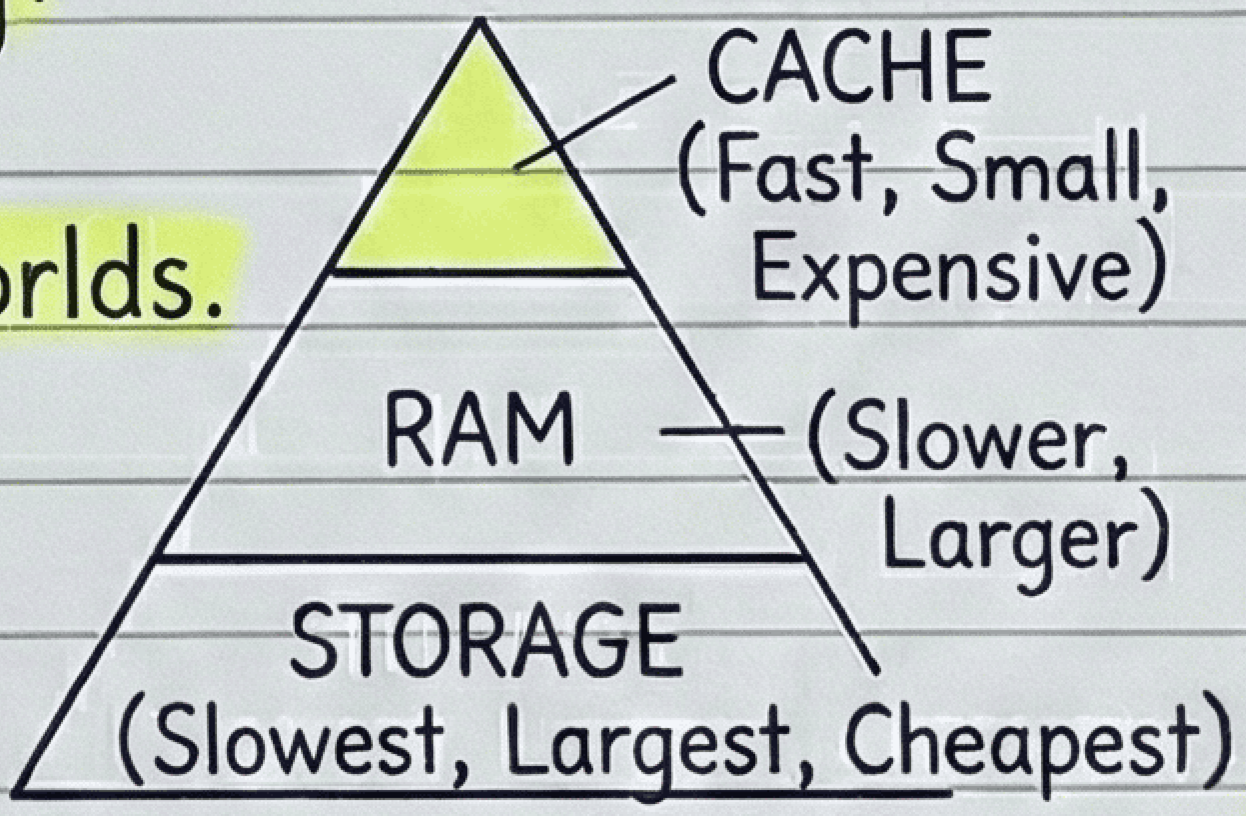
- Computers need memory that is very fast, very large, and very cheap, ALL at the same time.
- But that's not possible!

THE REALITY:

- Fast memory (like CPU caches) is expensive and small.
- Large memory (like hard drives/SSDs) is cheaper but slower.

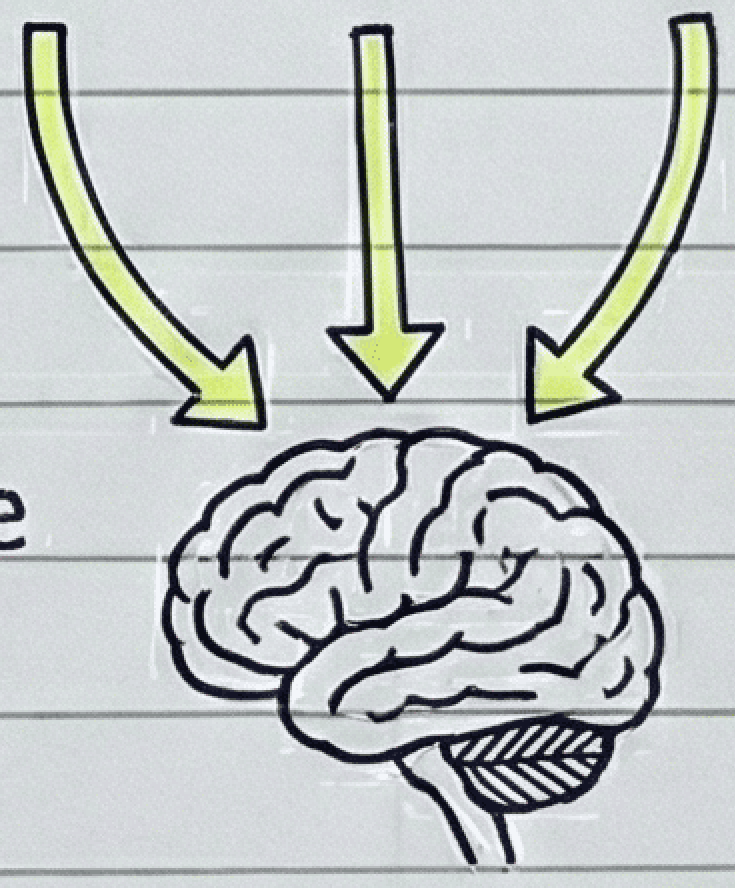


THE SOLUTION: Memory Hierarchy:



- A way to get the best of both worlds.
- Uses multiple levels of memory.

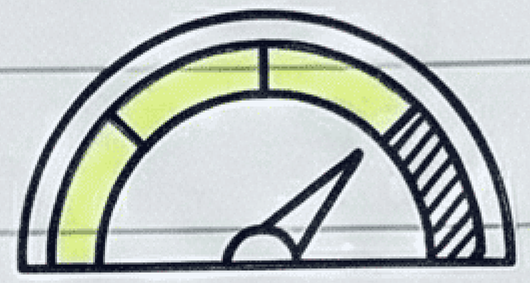
THE IDEA & BENEFIT:



- Keep the most frequently used data in the fastest memory levels.
- Balances speed, cost, and capacity.

Cache Memory

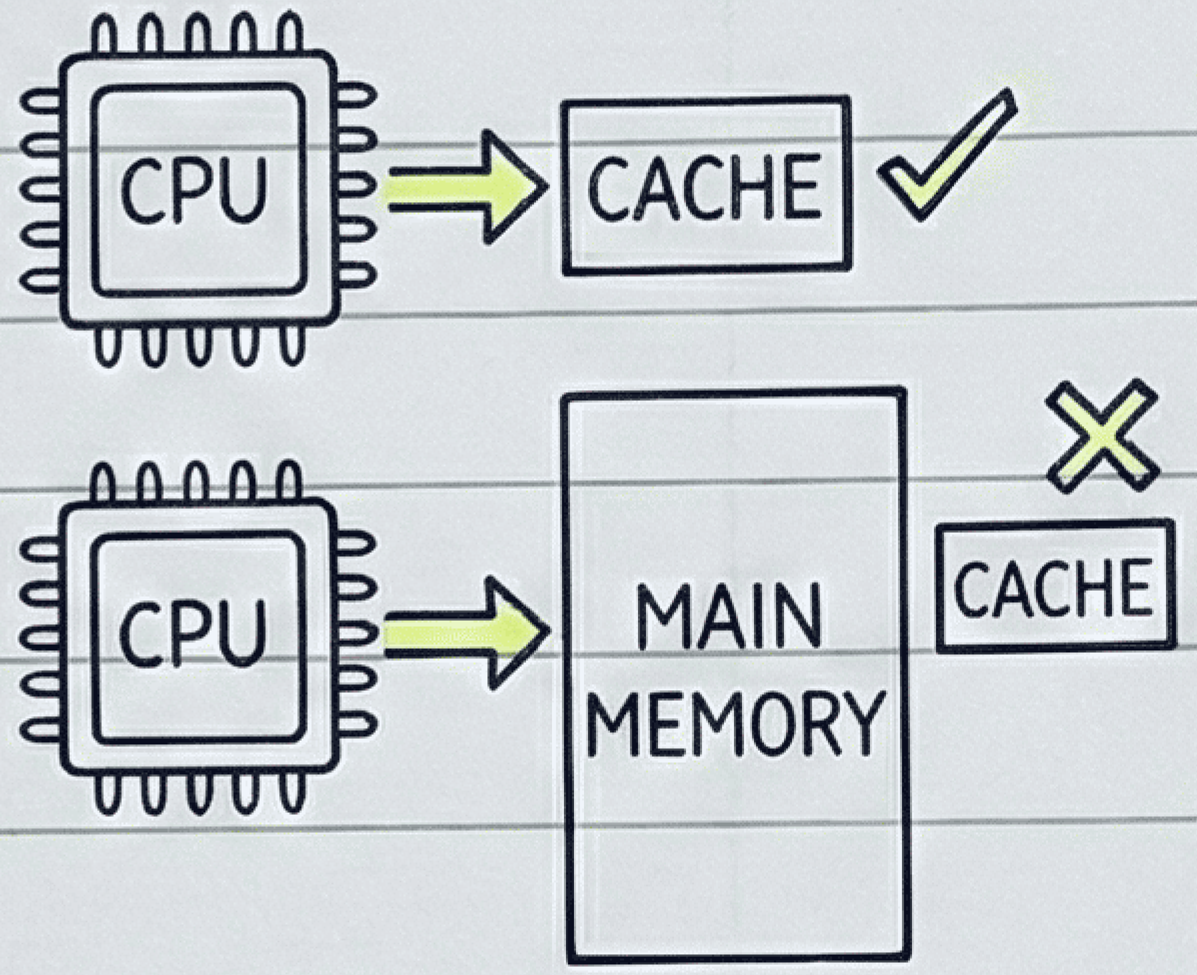
Performance Measurement: Hit Ratio



Performance is measured by hit ratio.

What are Hits & Misses?

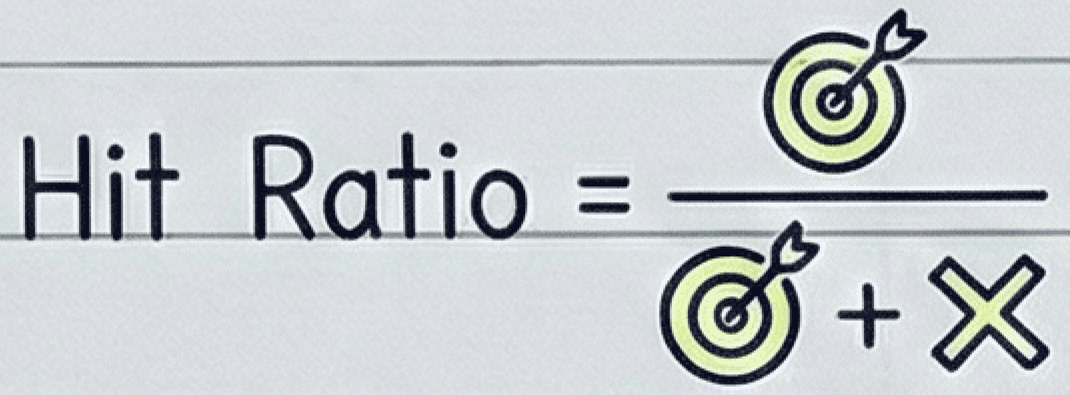
- CPU finds a word in cache → hit.
- Word not in cache, found in main memory → miss.



The Formula:

$$\text{Hit Ratio} = \frac{\text{Number of hits}}{\text{Total CPU references to memory}}$$

(Total references = hits + misses)



In Practice:

- Measured experimentally by running representative programs.
- Hit ratios of 0.9 and higher are common.
- This verifies the locality of reference property.

Mapping functions

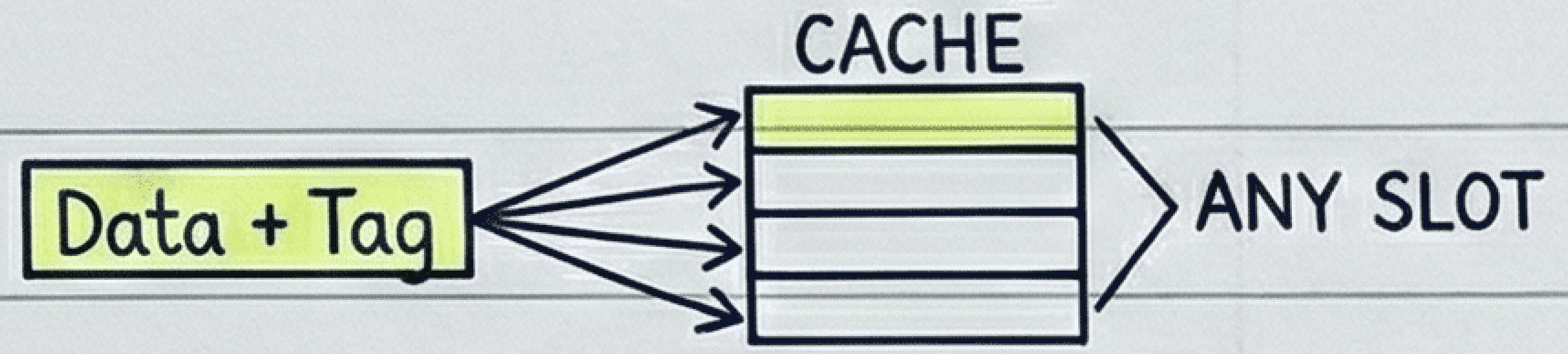
What is it?

The transformation of data from main memory to cache memory is referred to as a mapping process.

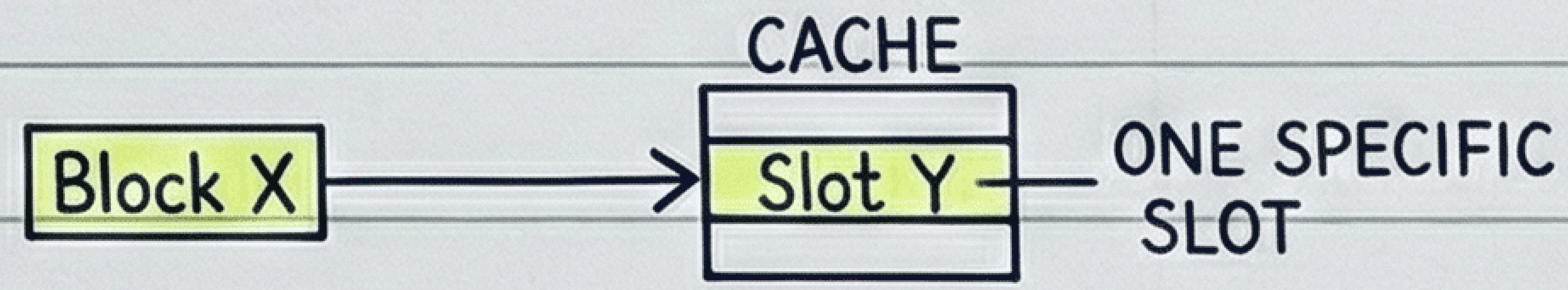
Types of Mapping Procedures:

Three types of mapping procedures are of practical interest when considering the organization of cache memory:

- Associative mapping



- Direct mapping

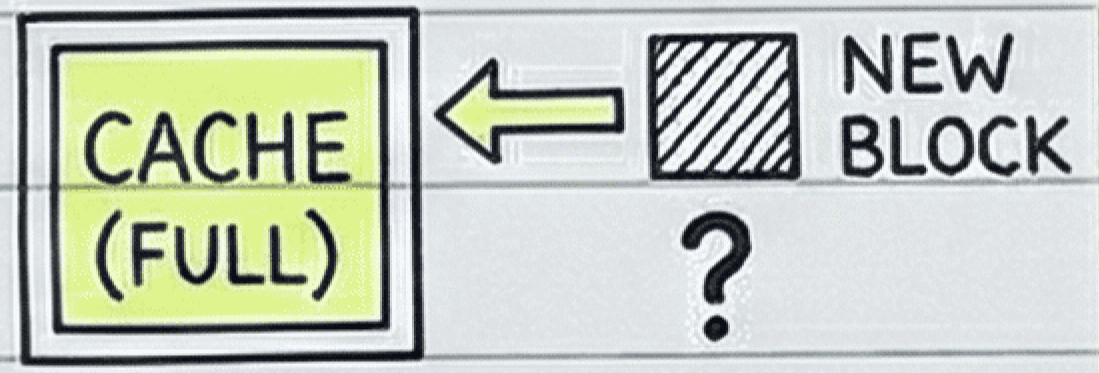


- Set-associative mapping

Replacement Algorithms

What are they?

When the cache is full, a decision must be made about which block to replace.

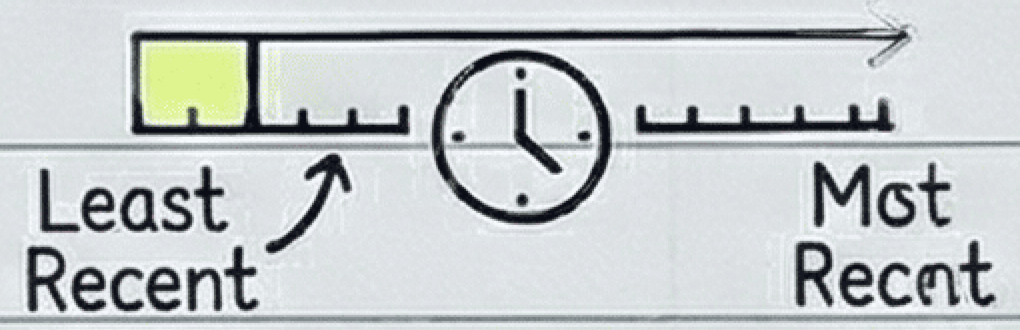


Common Algorithms:

- **Random replacement**: Chooses one tag-data item for replacement at random.

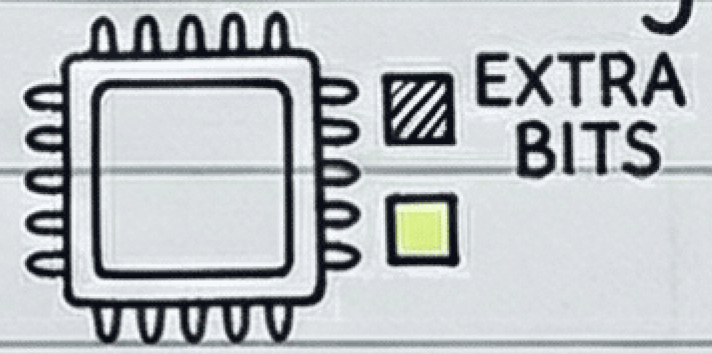
- **First-in, first-out (FIFO)**: Selects the item that has been in the set the longest.

- **Least recently used (LRU)**: Selects the item that has been least recently used by the CPU.



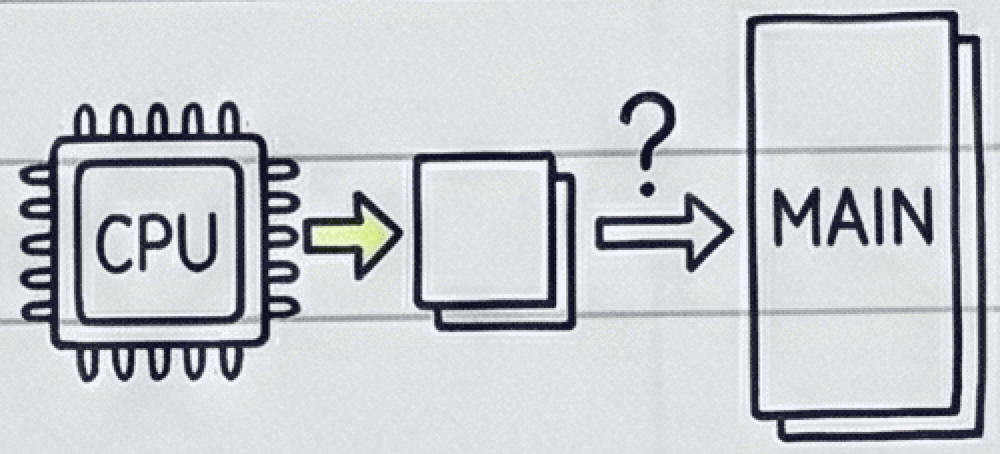
Notes on FIFO & LRU:

- FIFO is easy to implement, but can cause frequent removal/loading.
- LRU is more difficult to implement but is often better, assuming recent items are needed again.
- Both can be implemented by adding a few extra bits in each word of cache.



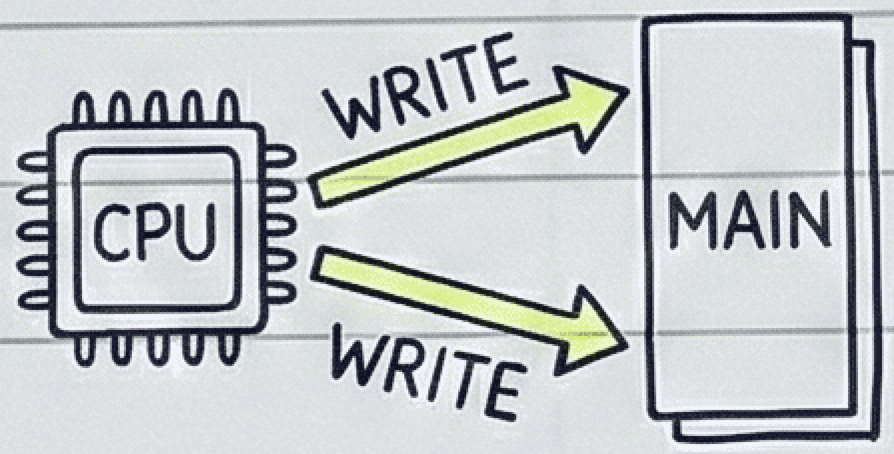
Write policies

The Issue:

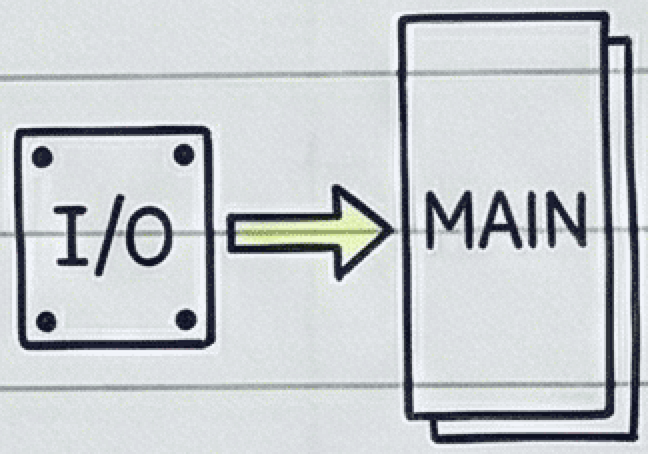


- An important aspect of cache is handling memory write requests when a word is in cache.

1. Write-through Method:

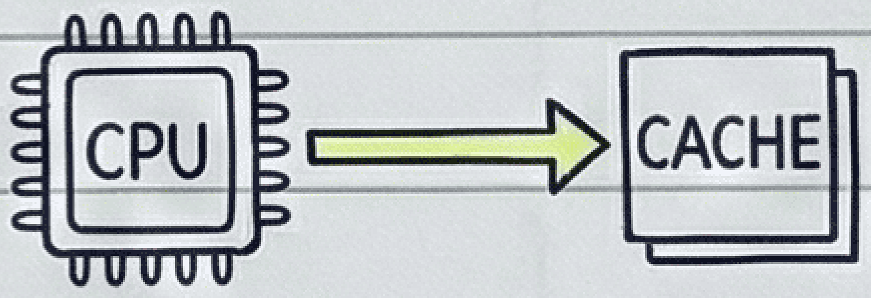


- Update main memory and cache memory in parallel with every write operation.

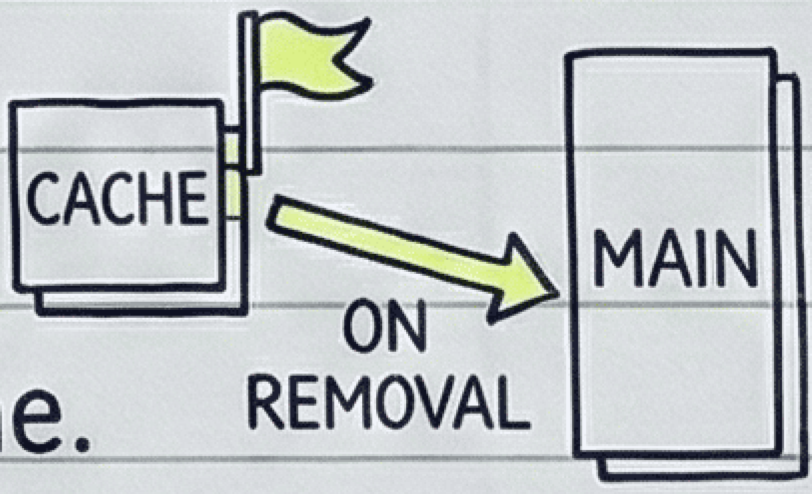


- **Advantage**: Main memory always has the same data as the cache. Important for Direct Memory Access (DMA) so I/O devices get the most recent data.

2. Write-back Method:



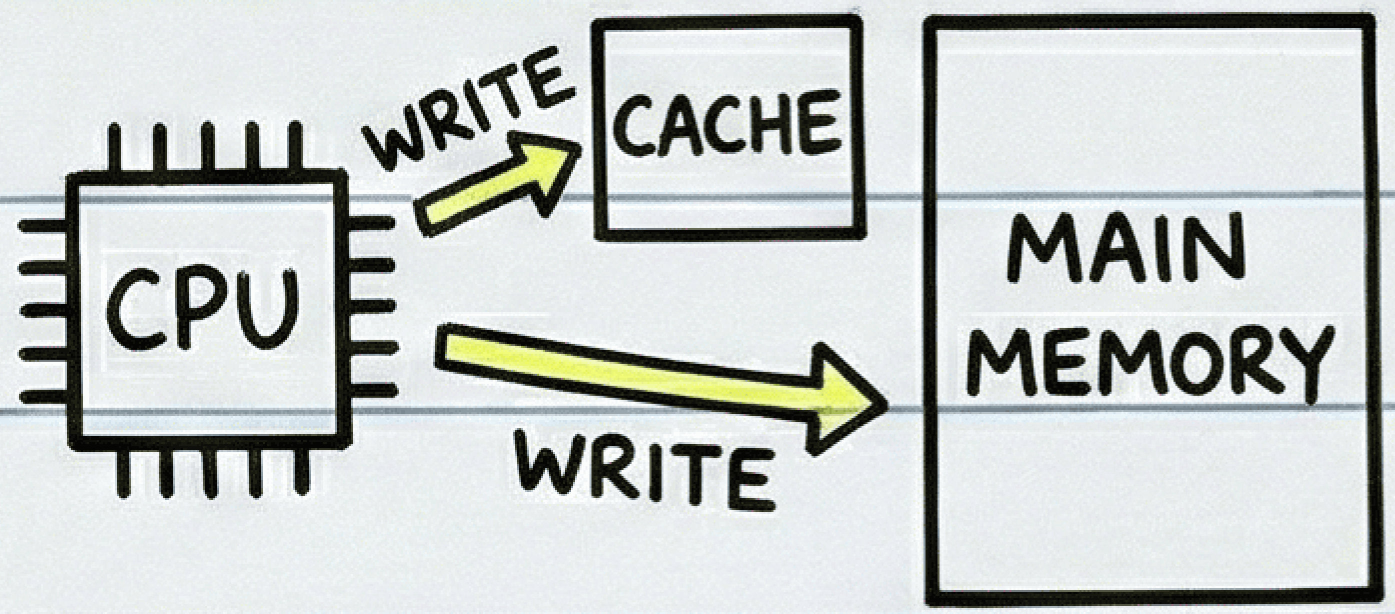
- Only the cache location is updated during a write.



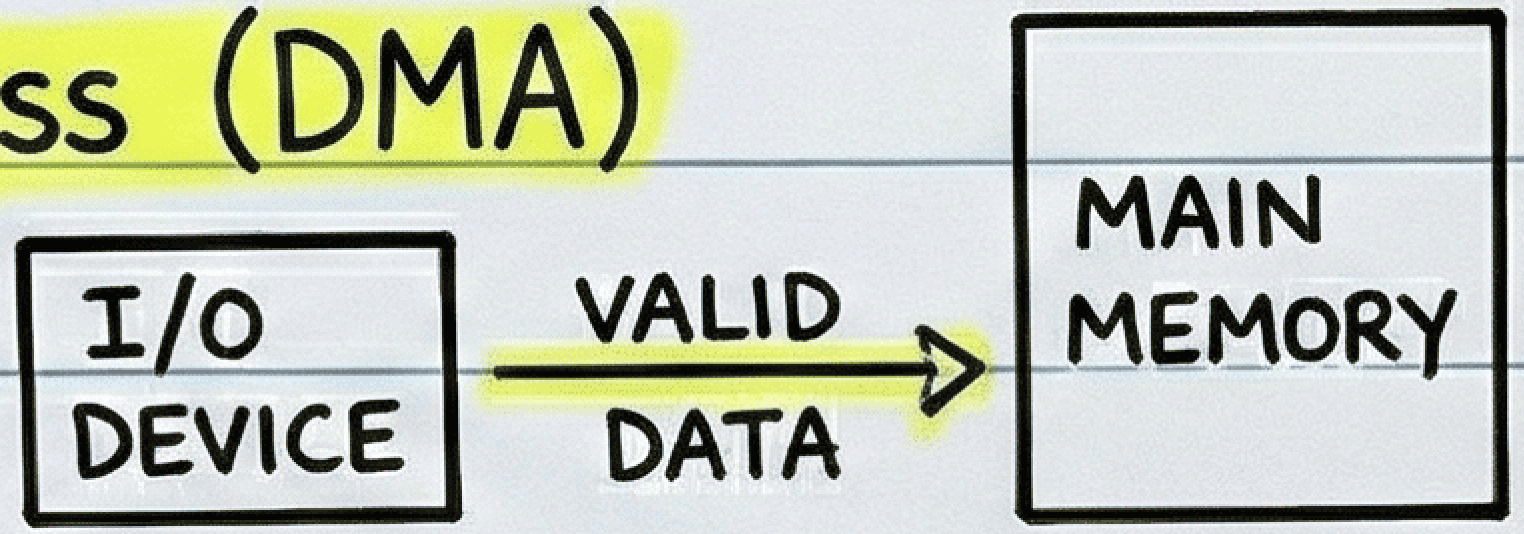
- The location is marked by a flag.
- The word is copied to main memory only when it is removed from the cache.
- **Reason**: A word might be updated multiple times while in cache.

- Memory writes are typically 10-30% of total memory references.

Advantage of Write-Through Cache Policy



- The main advantage is data consistency
- When the CPU writes, data is written to both the fast cache and the slower main memory at the same time.
- This ensures main memory always has the most up-to-date copy of the data.
- Important for systems with Direct Memory Access (DMA) or multiple devices.



- I/O devices will always see the correct data
- I/O devices will always see the correct data, and it is simpler to implement.